

8 Time Stretching Autotune

Goal: Change playback speed without changing the frequency content.

This is the same as changing pitch without changing the tempo. Same math.

If we change the sampling rate:

$$\mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$
$$\mathcal{F}\{x(at)\} = \int_{-\infty}^{\infty} x(at)e^{-j\omega t} dt$$

We can u-sub with $u = at$, $du = a dt$ to get

$$\mathcal{F}\{x(at)\} = \frac{1}{a} \int_{-\infty}^{\infty} x(u)e^{-j\frac{\omega}{a}u} du$$

This is just a dummy variable, so it really just looks like the Fourier transform but mapping frequencies from ω to $\frac{\omega}{a}$. This gives

$$\mathcal{F}(x(at)) = \frac{1}{a} X\left(j\frac{\omega}{a}\right)$$

When time is stretched by a factor a , each frequency ω gets mapped to $\frac{\omega}{a}$.

We have three possible approaches:

Overlap-Add (OLA) (bad approach)

1. We take a signal, DON'T do any Fourier. We break up the actual signal with a hopsize. We grab fixed analysis frames at each hop distance, with a length n .
2. Synthesis frames - we put the frames together with a different hop size. If the hop is smaller, then we go through the data faster. The analysis hop size is different from the synthesis frame. We have a stretching factor $= \frac{1}{a}$. Usually a is fixed, so we figure out what n to use by rearranging.

The new frame mapping is

$$y_n = \frac{n}{n - a}$$

The window must have a constant add (COLA condition from before).

3. Signal reconstruction

$$y_n = \frac{1}{n} \sum_{k=0}^{n-1} y_k$$

Pros and Cons

Cons

Can't preserve periodic structures - we are adding the music signal to itself, so if things don't line up then it completely changes the structures.

Transient doubling - If there are sharp transient signals that exist entirely within small portions of a window, then we might copy it multiple times when we make multiple frames. We can mitigate it with small N .

Pros

Preserves Timbre really well. This is good for percussive signals but not for pitches.

Instantaneous Frequency Estimation